Beyond Accuracy: How Parameter-Efficient Fine-Tuning and Prompting Interact in Movie Q&A

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Outline

1. The Problems
2. CinePile Dataset
3. Methodology
4. Key Findings
5. Next Steps

Problems

High Compute Cost

Full fine-tuning of LLMs demands massive memory and computation power – impractical for most research labs

Temporal and Hard Reasoning Gap

Base LLMs score only ~40% on temporal and Hard questions, struggling to track events across long movie narritives and answering difficult reasoning questions

Prompt Instability

Minor wording changes cause large accuracy swings – real world deployment risk for any MCQA system

Question Formation:

1. Which PEFT methods best adapts LLMs to long-form movie QA?
2. How do different prompt styles interact with fine-tuning?
3. Does fine-tuning reduce prompt sensitivity?

CINEPILE DATASET

305K
Questions

5+1
Reasoning Categories

4,941
Test Examples

TEMP
Temporal Reasoning
Timeline tracking across long narratives

CRD
Character & Dialogue
Speaker intent, relationships, interactions

NPA
Narrative & Plot
Story arc, cause-effect, plot sequence

STA
Setting & Atmosphere
Scene mood, location, visual context

TH
Theme Exploration
Abstract ideas, motifs, symbolism

HARD
Hard Split
Structurally complex multi-hop questions

Methodology: PEFT Method & Prompt Styles

Three Parameter-Efficient Fine-Tuning Methods

QLoRA

Weight-Space Modification

- Quantizes base weights to 4-bit
- Update the rank r =32. Higher r is better deep reasoning while not overfitting

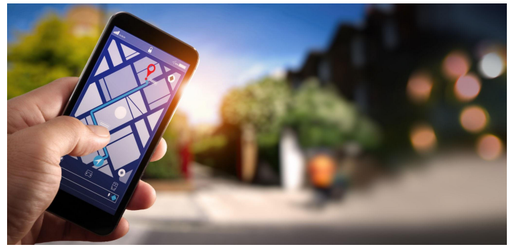
Prefix-Tuning

Virtual-Token Modification

- Tunes <0.1% parameters. 20-50 virtual tokens.
- Steers output format without touching weights.

ReFT

Representation-Space Intervention



- Low-rank edits to residual stream at layers 20, 24, 28.
- 10-50x fewer params than weight-based PEFT.

Three Prompt Styles (evaluated with 3 paraphrases each = 9 prompts total)

Base

Minimal instruction — just select the correct option.

Strict-Letter

Enforce single-letter answer (A–E) to prevent parsing errors.

Chain-of-Thought

Step-by-step reasoning before answering — boosts complex inference.

Key Findings

68.7%
QLoRA Accuracy
+8.7 pp over baseline

+15.7pp
Temporal Reasoning
Largest category gain

10×
Prompt Stability
Variance reduction with QLoRA

PEFT Method Comparison (Overall Accuracy %)

Prompt Style Insights

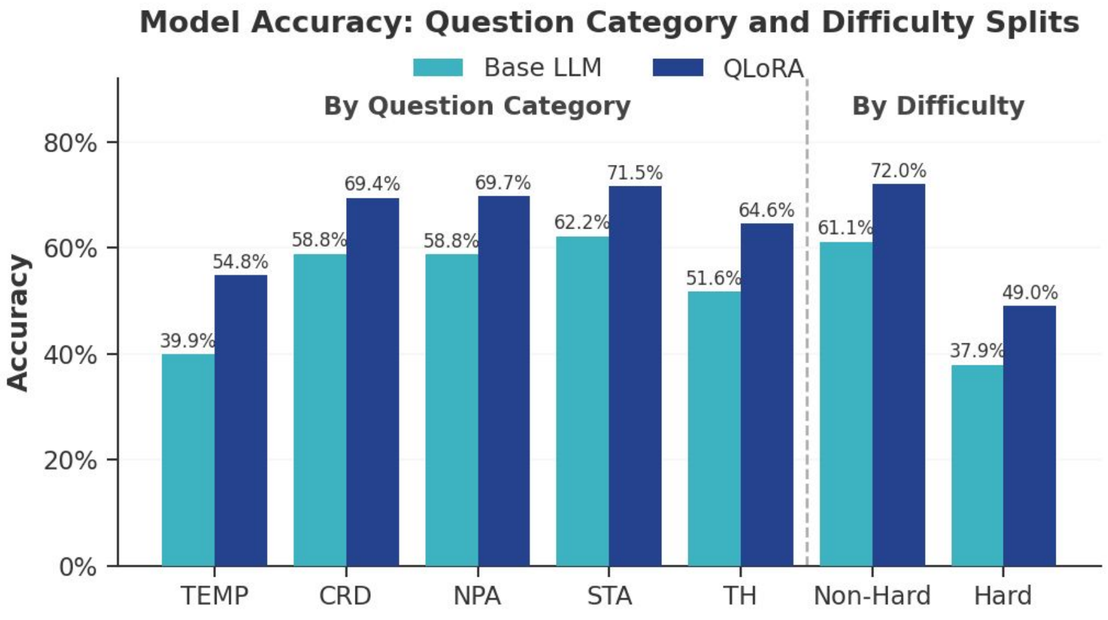
+ CoT boosts Temporal (+2.7%) and Theme (+2.6%)

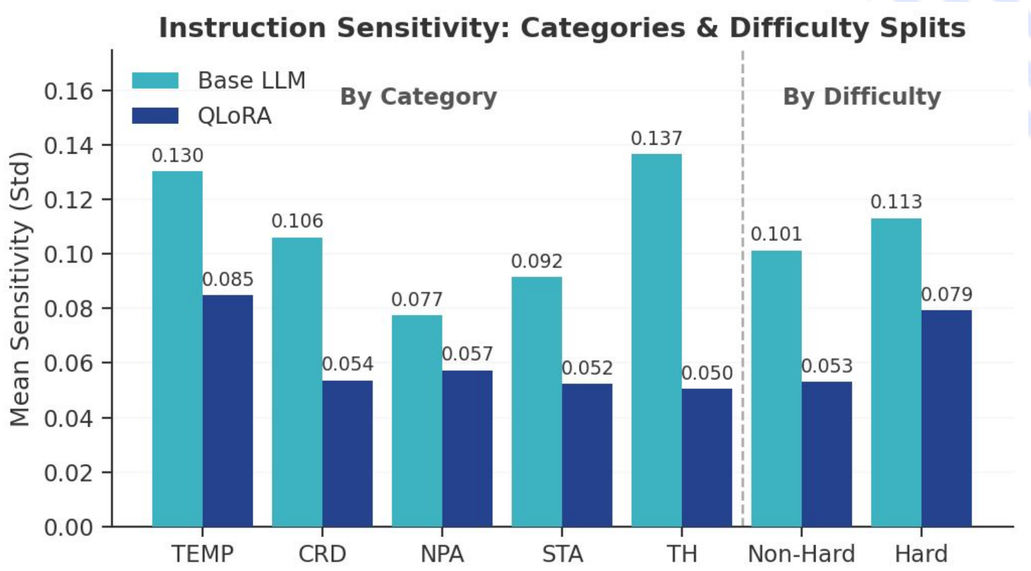
! CoT slightly hurts Narrative (NPA) — overstructures direct questions

~ Base model: high variance across paraphrases

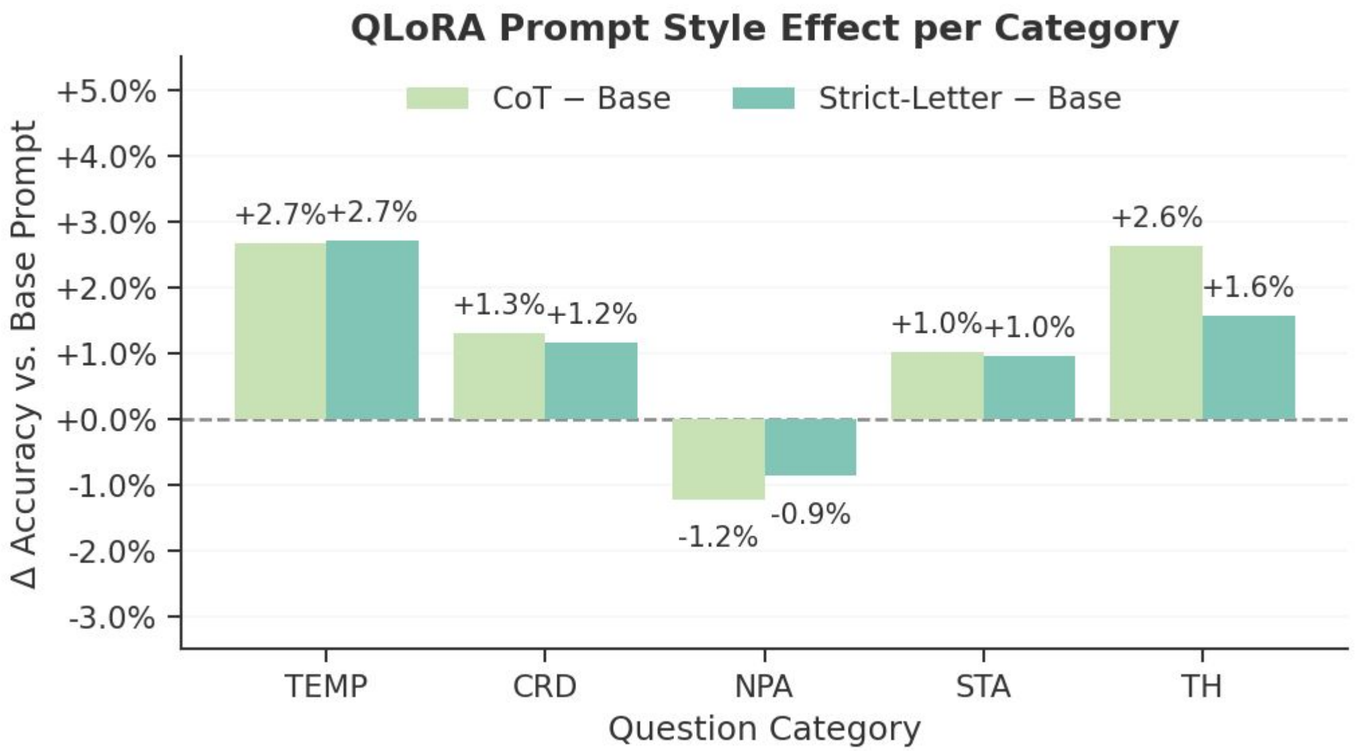
🛡️ QLoRA cuts prompt variance up to 10× — more stable for deployment

Key Findings





Key Findings



Next Steps

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01 Unquantized LoRA Evaluation

Test full-precision LoRA to isolate whether 4-bit quantization in QLoRA introduces accuracy loss on hard reasoning tasks.

02 Advanced Inference-Time Reasoning

Explore Tree of Thoughts (ToT) and Multi-Agent Debate to enable models to backtrack and self-correct multi-hop deductions.

03 Dynamic Prompt Routing

Train a lightweight classifier to detect question type at inference time and auto-select the optimal prompt style (CoT vs. Strict-Letter).

04 Vision-Language Models (VLMs)

Apply PEFT interventions to multimodal VLMs — grounded visual embeddings may resolve narrative ambiguities text-only models still face.

The End

Thank You